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Tectonic elements with South China affinity in the Korean Peninsula: new Early Jurassic palaeomagnetic data

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SUMMARY

Early Jurassic sandstones and shales were collected at 14 sites from the Kimpo area in the western part of the Korean Peninsula for a palaeomagnetic study. A characteristic remanent magnetization component, which passes the fold test, was isolated from seven sites. This characteristic component is interpreted to be of primary Early Jurassic origin because the folding age is Middle–Late Jurassic. The primary directions, together with previously reported ones, provide a mean direction ($D = 38.1^\circ$, $I = 39.7^\circ$, $\alpha_{95} = 3.8^\circ$, 11 sites) and a palaeomagnetic pole (54.1°N, 230.2°E, $A_{95} = 4.1^\circ$) for the Kimpo area. This palaeomagnetic pole shows good agreement with the coeval poles for the South China Block and is significantly removed from that for the North China Block, suggesting that the Kimpo area has been part of the South China Block since at least Early Jurassic times. The Kimpo area constitutes a part of the Imjingang Belt which is palaeontologically allocated to the South China Block in its origin, while all Precambrian tectonic blocks that compose the basement of the Korean Peninsula are interpreted to be of North China origin. We conclude that the Imjingang Belt, including the Kimpo area, is a tectonic extension of the South China Block on the Korean Peninsula. The South China elements were obducted onto the North China Block at the site of the present Korean Peninsula during the collision event of the North and South China Blocks, and hence the Korean Peninsula is composed of areas with both North and South China affinities.

Key words: Chinese blocks, Korea, palaeomagnetism, tectonics.

1 INTRODUCTION

The collision between the North and South China Blocks is one of the most prominent tectonic events in East Asia since Late Palaeozoic times. The South China Block began to collide with the North China Block between the Permian and Triassic, and the collision ceased during the Middle–Late Jurassic (Gilder & Courtillot 1997) or the Early Cretaceous (Yokoyama *et al.* 2001). The collision between the two blocks occurred initially in their eastern parts, and the suturing progressed westwards (Zhao & Coe 1987). The North and South China Blocks were united following a scissor-like collision process that included counter-clockwise rotation of the North China Block and clockwise rotation of the South China Block (Zhao & Coe 1987).

The Korean Peninsula is situated to the east of the North and South China Blocks (Fig. 1a). Analysis of fossils shows that it is palaeontologically composed of fauna with both North and South China affinities (Kobayashi 1966, 1967). It thus appears that the eastern extension of the boundary between the two blocks passes through the Korean Peninsula. Several authors have made attempts to delineate the geological boundaries between the North and South China Blocks within the Korean Peninsula (Ree *et al.* 1996; Zhang 1997; Chough *et al.* 2000). Although they agreed that the Korean Peninsula was involved in the collision event between the North and South China Blocks at their eastern parts, the proposed boundaries are not consistent.

Palaeomagnetic pole positions from the Korean Peninsula can provide a basis for recognizing its affinity with the Chinese blocks. Because Early Mesozoic palaeomagnetic poles from the North and South China Blocks are significantly removed from each other (Enkin *et al.* 1992; Zhao *et al.* 1996; Gilder & Courtillot 1997; Yang & Besse 2001), a locality on the Korean Peninsula yielding an Early Mesozoic palaeomagnetic pole can be allocated to either of the Chinese blocks in terms of location of poles.

This study focuses on the tectonics of the Korean Peninsula using palaeomagnetic evidence from the Early Jurassic Munsusan (Munjusan) Formation in the Daedong (Taedong) Supergroup, with the objective of comparing the palaeomagnetic pole from this formation with those from the North and South China Blocks. Because relative rotation of 10–30° between the North and South China Blocks occurred after the Early Jurassic (Gilder *et al.* 1993; Uno & Huang 2003), the Munsusan Formation is expected to record tectonic rotations associated with the collision between these two blocks. The

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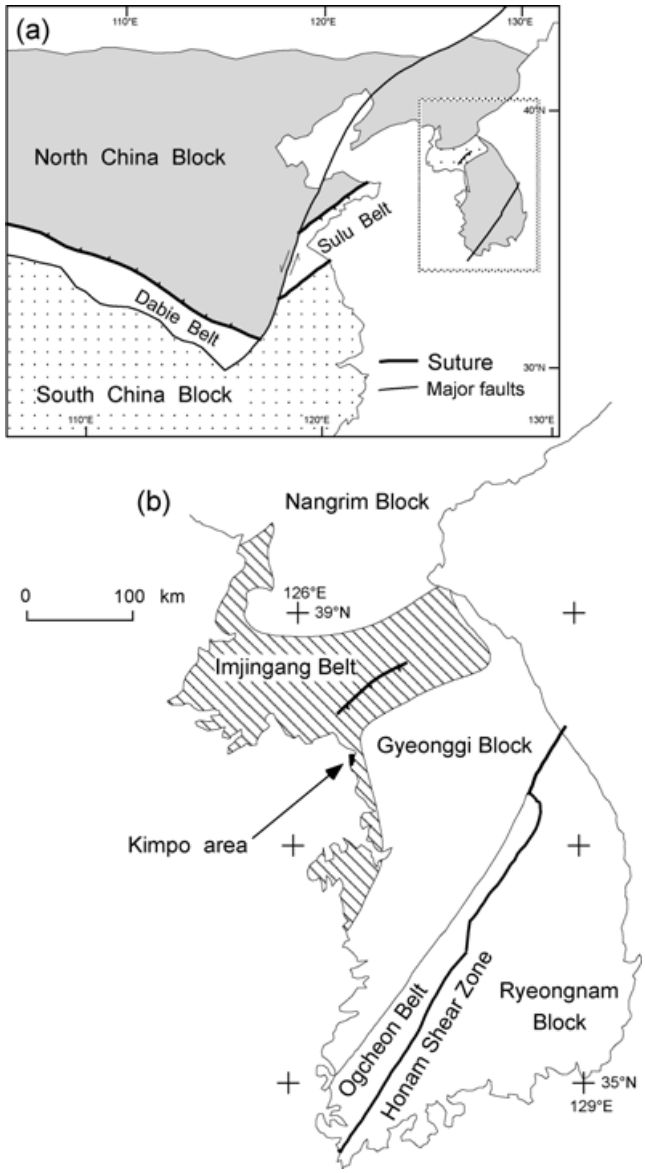


Figure 1. (a) Outline tectonic map of the Korean Peninsula and adjacent area. (b) Major tectonic divisions of the Korean Peninsula. The distribution area of the Imjingang Belt is after Otoh *et al.* (1999).

study area is in the Kimpo area (Fig. 1b), where a fold test is available and no large faults occur (Reedman & Um 1975; Lee 1987).

There have been two studies of Jurassic rocks around the study area (Kim & Van der Voo 1990; Kim *et al.* 1993); however, their tectonic interpretations are not consistent. Moreover, the tectonic interpretations of the two studies are not based on the notion that the Korean Peninsula is a collage of tectonic elements of both North and South China affinities. This paper provides a reliable view on the division of the Korean Peninsula between the North and South China Blocks.

2 GEOLOGICAL SETTING AND SAMPLING

The Korean Peninsula is composed of three tectonic divisions with Precambrian basements, namely the Nangrim, Gyeonggi and Ryeongnam Blocks from north to south (Reedman & Um 1975; Cluzel *et al.* 1990) (Fig. 1b). The former two blocks are bounded by the Imjingang Belt, and the latter two by the Ogcheon Belt.

The Late Triassic–Early Jurassic Daedong Supergroup is sporadically distributed in the Korean Peninsula. This supergroup is mostly observed in the northern and western margin of the Imjingang Belt (Otoh & Sasaki 1998). The Daedong Supergroup is well exposed in the Kimpo area in western part of the Korean Peninsula (Fig. 1b). It has a total thickness of about 1600 m and is divided into two groups: the Tongjin and Munsusan formations in ascending order (Reedman & Um 1975; Lee 1987). The former formation is made up of alternating sandstones and shales, and the latter formation is composed of shales, sandstones and conglomerates. The age of the Tongjin Formation is determined as Late Triassic based on the occurrence of plant fossils of *Anthrophyopsis*, *Drepanozamites* and *Cycadocarpidium* (Kimura & Kim 1984). The Munsusan Formation, the upper part of the supergroup, is assigned an Early Jurassic age on the basis of stratigraphic and structural data (Chun *et al.* 1988).

The folds of the Daedong Supergroup in the Kimpo area were formed during the Daebo Orogeny (Reedman & Um 1975; Lee 1987). The Daebo Orogeny occurred after the deposition of the Daedong Supergroup and ceased with the deposition of the Early Cretaceous Gyeongsang Supergroup. Hence the Middle–Late Jurassic age of the folding is established (Cluzel & Cadet 1992). The fold axes in the Kimpo area generally trend northeast–southwest in its central and southern parts but ENE–WSW in the northern part (Thomas & Park 1976). The granites were emplaced into the southern and eastern parts of the Kimpo area during the Daebo Orogeny.

One hundred and thirteen hand samples of the Daedong Supergroup were collected at 14 sites from the Munsusan Formation for palaeomagnetic analysis (Fig. 2). The samples consist of red and grey sandstones and red shales (Table 1). The sampling sites show monoclinal structures with dips ranging from 17 to 63°. In each site, eight to nine hand samples were oriented by magnetic compass.

3 PALAEOMAGNETISM

In the laboratory, cores of diameter 25 mm and length 22 mm were cored from each hand sample. Natural remanent magnetizations (NRMs) were measured with a 2-G superconducting rock magnetometer (SRM) in a magnetically shielded room at Kyoto University. All samples were subjected to progressive thermal demagnetization treatment using a Natsuhara TDS-1 thermal demagnetizer. Results for each sample were plotted on orthogonal vector diagrams (Zijderveld 1967) to evaluate their demagnetization behaviours. Principal component analysis (Kirschvink 1980) was used to estimate the directions of the observed magnetic components (Table 1). Palaeomagnetic data were analysed using the application software of Cogné (2003). In addition, the anisotropy of magnetic susceptibility (AMS) of the samples was measured with a Kappabridge susceptibility meter KLY-3.

Remanent magnetizations were divided into two types based on their inclination values relative to bedding planes. At seven sites (MJ2, 3, 4, 5, 6, 7 and 8, Figs 3a to e), a magnetization component with northeasterly declinations and intermediate inclinations after tilt correction was isolated (here referred to as type A). At four sites (MJ9, 10, 11 and 12, Figs 3f and g), a magnetization component with northeasterly declinations and shallow inclinations after tilt correction was observed (referred to as type B). The remaining three sites exhibit either erratic demagnetization behaviour at

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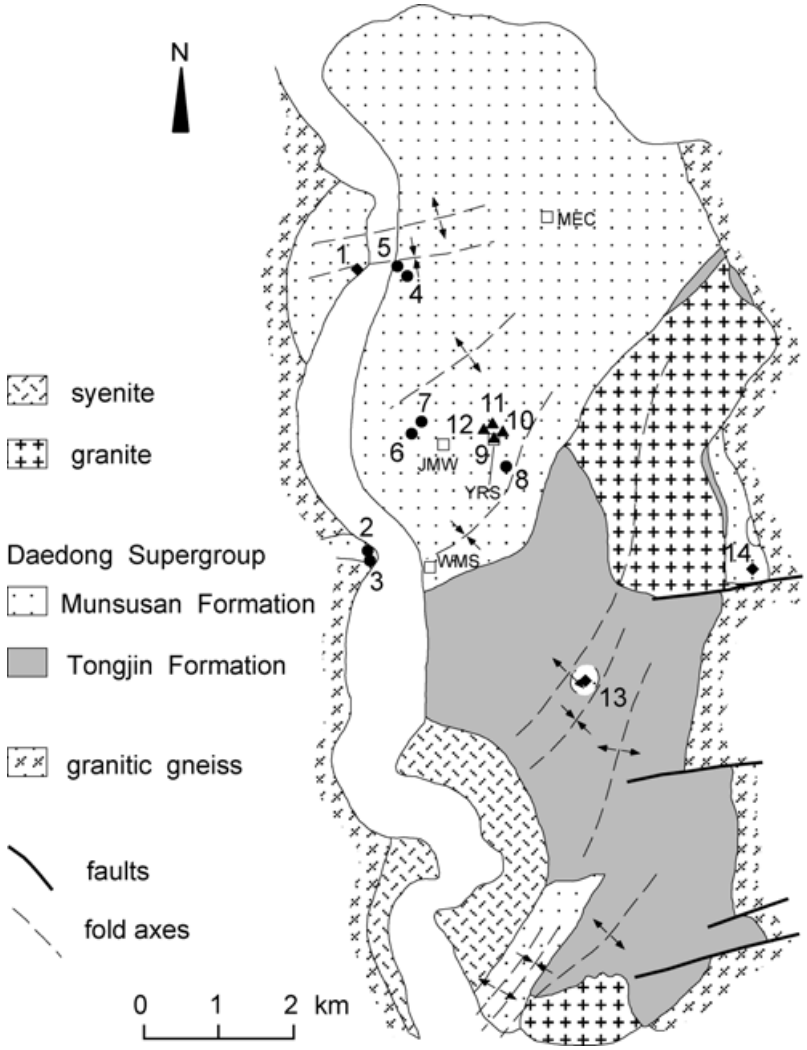


Figure 2. Geological map of the Kimpo area. Solid symbols denote the palaeomagnetic sampling sites of this study: circles, sites with characteristic palaeomagnetic directions; triangles, sites with low inclination values; diamonds, sites with erratic or inconsistent directions. Open squares are sites from which palaeomagnetic results are described by Kim *et al.* (1993).

higher temperatures (MJ1 and 14, Fig. 3h) or inconsistent directions ($\alpha_{95} > 25^\circ$) of remanent magnetization within a site (MJ13, Table 1).

3.1 Type A (sites MJ2, 3, 4, 5, 6, 7 and 8)

A characteristic remanent magnetization component of sites MJ2 and 3 shows unblocking temperatures ranging from 670 to 690 °C, and more than half of the samples have an unblocking temperature of 680 °C. The large decay in remanent magnetization intensity between 630 and 670–690 °C suggests that haematite is the predominant carrier of the remanent magnetization.

The characteristic component of sites MJ4–8 is resolved in the form of higher-temperature trajectories and generally has unblocking temperatures of 500–580 °C. Some samples display unblocking temperatures above 580 °C; however, a large fraction of this component is demagnetized by 580 °C, and the remanence is completely demagnetized by 650 °C (Fig. 3d). The unblocking temperatures indicate that the characteristic component resides in either magnetite or magnetite and haematite.

The characteristic component from six sites shows normal polarity, while this component from one site contains a reversed magnetic polarity: site MJ6 is predominantly of normal polarity and has a single hand sample showing reversed polarity (Fig. 3e). This aspect of site MJ6 is stratigraphically related. The sample of reversed polarity is stratigraphically situated higher in the sequence than those of normal polarity. The reversal test has no statistical significance because there is only one direction in the population of reversed polarity, although the direction of reversed polarity is roughly close to those of normal polarity.

The predominance of normal polarity during Hettangian times (earliest Jurassic) has been revealed from the magnetostratigraphic sequence of Newark Basin, eastern North America (Kent *et al.* 1995), the Colorado Plateau, western North America (Molina-Garza *et al.* 2003) and the Paris Basin, western Europe (Yang *et al.* 1996). These magnetostratigraphic records of earliest Jurassic times exhibit predominantly normal polarity divided by short reversed polarity zones. Since the characteristic direction of the present study is almost entirely of normal polarity, the bulk of the characteristic direction may be interpreted as a record of this period.

The seven site-mean directions of the characteristic component have precision parameters (k value) ranging from 20.6 to 502.5 (Table 1). These directions before tilt correction, with a mean of

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Table 1. Palaeomagnetic results of the Munsusan Formation.

Site	Lithology	N	*In situ*		Tilt corrected				Locality				
			D (°)	I (°)	D (°)	I (°)	k	α_{95} (°)	Lat. (°N)	Long. (°E)	Strike (°)	Dip (°)	DD (°)
MJ1	Grey ss								37.739	126.525	275	37	5
MJ2	Red shale	8	27.5	52.4	30.7	35.8	451.7	2.6	37.703	126.522	310	17	40
MJ3	Red shale	8	51.0	56.5	34.8	34.2	502.5	2.5	37.702	126.522	277	27	7
MJ4	Grey ss	5	65.1	55.5	36.9	45.1	20.6	17.3	37.739	126.530	254	25	344
MJ5	Grey ss	6	41.5	52.5	35.8	40.6	61.2	8.6	37.739	126.529	284	13	14
MJ6	Grey ss	5	22.7	34.4	40.0	45.1	58.0	10.1	37.717	126.532	58	23	148
MJ7	Grey ss	5	15.2	54.1	43.7	43.9	59.3	10.0	37.718	126.532	7	26	97
MJ8	Grey ss	3	31.7	59.0	32.7	38.0	346.9	6.6	37.712	126.548	304	21	34
MJ9[1]	Red ss	9	27.9	57.1	47.0	14.8	493.6	2.3	37.715	126.544	338	48	68
MJ10[1]	Red ss	6	27.1	56.9	42.4	25.7	644.7	2.6	37.716	126.544	333	35	63
MJ11[1]	Red ss	7	32.7	51.0	45.9	19.4	814.7	2.1	37.716	126.542	339	36	69
MJ12[1]	Red ss	9	36.3	49.4	48.7	10.9	267.9	3.2	37.716	126.542	340	43	70
MJ13[2]	Red ss	8	61.2	68.2	96.6	12.8	4.9	28.0	37.688	126.557	24	63	114
MJ14	Grey ss								37.699	126.584	300	48	30
Mean		7	35.3	53.2			38.0	9.9	37.700	126.500			
					36.2	40.5	209.6	4.2					
Kim et al. (1993)													
MEC		10			47.8	36.0	27.8	13.9					
JMW		7			52.8	39.4	11.0	19.1					
WMS		8			31.9	37.5	29.7	10.3					
YRS[1]		13			43.0	10.8	13.0	8.0					
Kim & Van der Voo (1990)													
Jmj		4	18.9	34.0	33.0	39.3	102.0	9.2					
J1 Mean		11			38.1	39.7	145.6	3.8	37.700	126.500			

Symbols and abbreviations: ss, sandstone; N, the number of the samples in calculation for the means; D, declination; I, inclination; k, the precision parameter; α_{95}, radius of the cone of 95 per cent confidence; Lat., north latitude; Long., east longitude; DD, dip direction.
The Early Jurassic (J1) mean direction is calculated by combining the present data with those of Kim *et al.* (1993) and Kim & Van der Voo (1990).
[1]Not involved in the calculation of the mean direction due to anomalous inclination values.
[2]Not involved in the calculation of the mean direction due to a site $\alpha_{95} > 25°$.

$D_g = 35.3°$, $I_g = 53.2°$ with $\alpha_{95} = 9.9°$, are significantly distinct from the present Earth's field direction ($D = 352.4°$, $I = 54.4°$). A mean direction of the seven sites after tilt correction is $D_s = 36.2°$, $I_s = 40.5°$ with $\alpha_{95} = 4.2°$ and reveals a tight clustering (Fig. 4a). Tilt correction increases the precision parameter from 38.0 to 209.6, and a fold test is positive at the 95 per cent confidence level (McFadden 1990), indicating that the characteristic component is of pre-folding origin.

The observed palaeomagnetic direction is characterized by easterly deflected declinations and similar to Cretaceous expected directions from the Korean Peninsula with declinations of 24–38° (Lee *et al.* 1987; Zhao *et al.* 1999; Doh *et al.* 2002). The fold test shows that the remanent magnetization was acquired prior to folding, and the folding age of Middle–Late Jurassic is established on the basis of the previously discussed stratigraphic and structural data (Cluzel & Cadet 1992). We interpret these observations to show that the observed direction is not attributed to a secondary magnetization and is essentially of primary origin.

The primary directions are observed at the seven sites in the Kimpo area. We determine an Early Jurassic palaeomagnetic direction for the Kimpo area based on data sets from the present and previous studies (Kim & Van der Voo 1990; Kim *et al.* 1993). Kim *et al.* (1993) described three site-mean directions of both normal and reversed polarities, with α_{95} values ranging between 10.3° and 19.1° (sites MEC, JMW and WMS, Fig. 2). Kim & Van der Voo (1990) presented one site-mean direction of normal polarity from the Kimpo area (site Jmj; the precise location of the site is not described). Since the directions of the previous studies are statistically coincident with the primary directions of the present study (Fig. 4a), palaeomagnetic directions from the three studies are combined to yielded a mean direction of $D = 38.1°$, $I = 39.7°$ with $\alpha_{95} = 3.8°$ at the reference point of the Kimpo area (37.7°N, 126.5°E). We regard this mean direction as the representative of the Kimpo area in the Early Jurassic.

3.2 Type B (sites MJ9, 10, 11 and 12)

A characteristic component of these sites shows a steady decay in intensity during thermal demagnetization. The most commonly observed unblocking temperatures of the characteristic component are between 650 and 660 °C, suggesting that the remanence predominantly resides in haematite.

The four site-mean directions of the characteristic component cluster well at each site, with the precision parameter ranging from 267.9 to 814.7 (Table 1). Mean directions of the four sites before and after tilt correction are $D_g = 31.3°$, $I_g = 53.7°$ with $\alpha_{95} = 5.4°$ and $D_s = 46.1°$, $I_s = 17.7°$ with $\alpha_{95} = 7.8°$ respectively. The tilt-corrected mean direction is clearly distinct from the primary direction of type A. This disagreement can be explained in terms of difference in inclination, because the declinations are relatively consistent between types A and B. The mean direction of type B, after tilt correction, is shallower by $22.1 \pm 8.9°$ than that of type A, with no significant declination contrast of $7.5 \pm 9.8°$.

The shallow inclination is observed only from neighbouring sites in an area (sites MJ9–12, Fig. 2). Observations of the previous results

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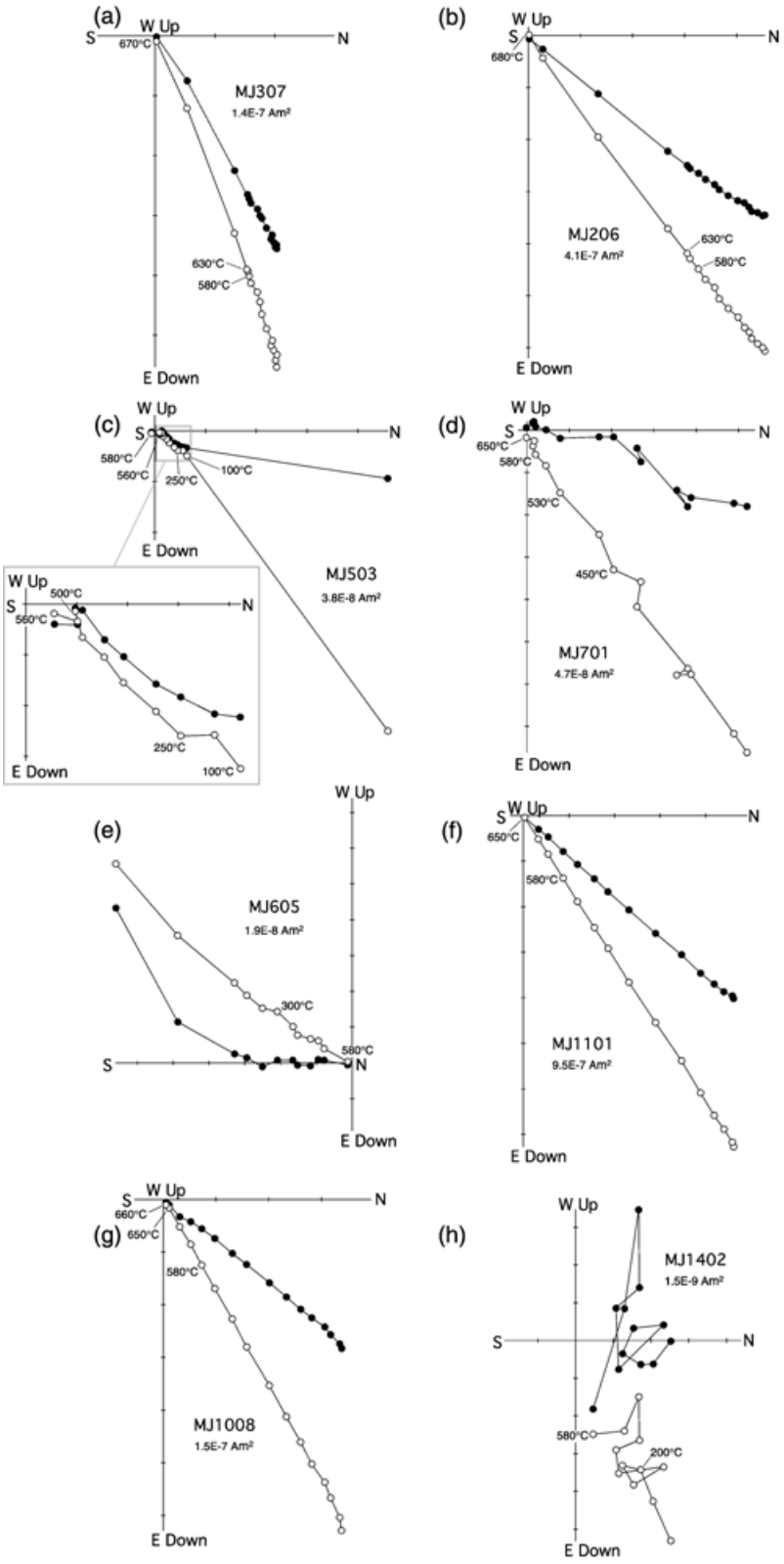


Figure 3. Orthogonal plots of thermal demagnetization data before tilt correction. Solid and open symbols in the demagnetization diagrams represent projections on horizontal and vertical planes, respectively.

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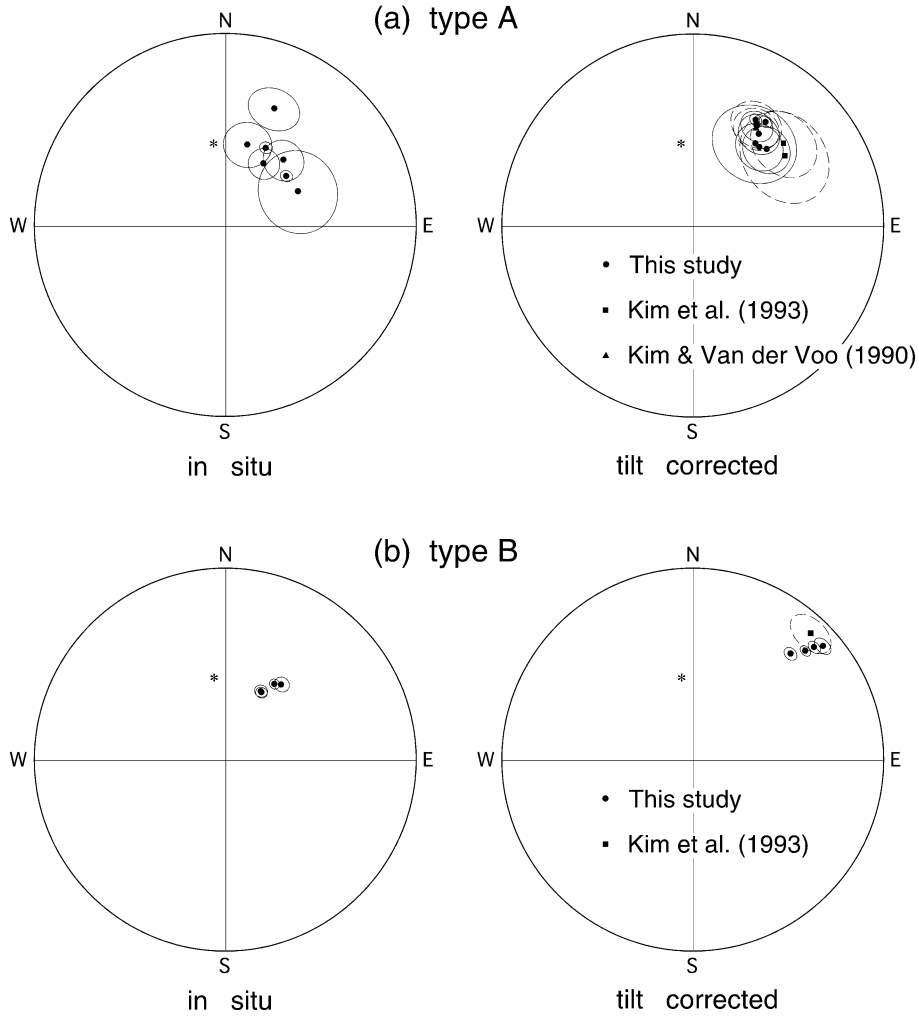


Figure 4. (a) Equal area projections of directions of the characteristic component (solid circles, with associated 95 per cent confidence limits) from seven site of the Munsusan Formation (type A). The tilt-corrected directions (right) are drawn with those from three sites of the same formation by Kim *et al.* (1993) (solid squares, with associated 95 per cent confidence limits in dashed lines) and one site by Kim & Van der Voo (1990) (solid triangle, with associated 95 per cent confidence limits in dashed lines). (b) Equal area projections of directions with anomalously low inclinations (solid circles, with associated 95 per cent confidence limits) from four sites of the Munsusan Formation (type B). The tilt-corrected directions (right) are drawn with a tilt-corrected direction from the same area by Kim *et al.* (1993) (solid squares, with associated 95 per cent confidence limits in dashed lines). The asterisk denotes the present Earth's field direction.

corroborate that the presence of samples with shallow inclination is confined to a small area. Kim *et al.* (1993) reported directions with shallow inclinations of about 10° from a single site (YRS, Fig. 2). This site is located in the area incorporating sites MJ9–12 of the present study. A comparison of the directions between the present study and previous one is made in Fig. 4b and shows good agreement.

AMS measurements were performed in order to compare directions between remanence and the maximum axis of magnetic susceptibility and to compare magnetic fabrics of samples between types A and B. AMS results suggest that the shallow inclination in sites MJ9–12 (type B) mirrors the magnetic fabrics in the samples. The maximum axes of samples of type B dip at low angles of about 15° in northeasterly directions (Fig. 5a). These directions are similar, essentially in inclination, to the remanence directions of type B. In contrast to the case for type B, samples of type A show that the maximum axes are dispersed in the stratigraphic horizontal plane, although the distribution of the axes appears to be restricted to northerly directions. The shape parameter (T) and the degree of anisotropy (P') are compared (Jelínek 1981) between samples of types A and B. Although the difference in the P' values between types A and B are not quite demonstrably significant, the T values for type B display scattering in the area corresponding to the oblate fabrics and those for type A show more neutral patterns (Fig. 5b). This observation shows that the dominant feature of type B is the oblate AMS ellipsoid with the shallow dips of the maximum axes, and therefore that the inclinations observed in type B are interpreted to be affected by the magnetic fabrics.

It should be noted again that the type B remanence is found only from a restricted area incorporating sites MJ9–12 (Fig. 2). This aspect implies that inclination shallowing is a localized phenomenon. This notion is supported by the fact that the type B remanence of the previous study (site YRS, Kim *et al.* 1993) is also observed from the same area containing sites MJ9–12 of the present study. Geological observations of this area provide a basis for explaining the shallowing effect. According to Thomas & Park (1976), sandstone layers in the sampling locality show cross-laminations, which form as a result of deposition from currents. Such a sedimentary environment

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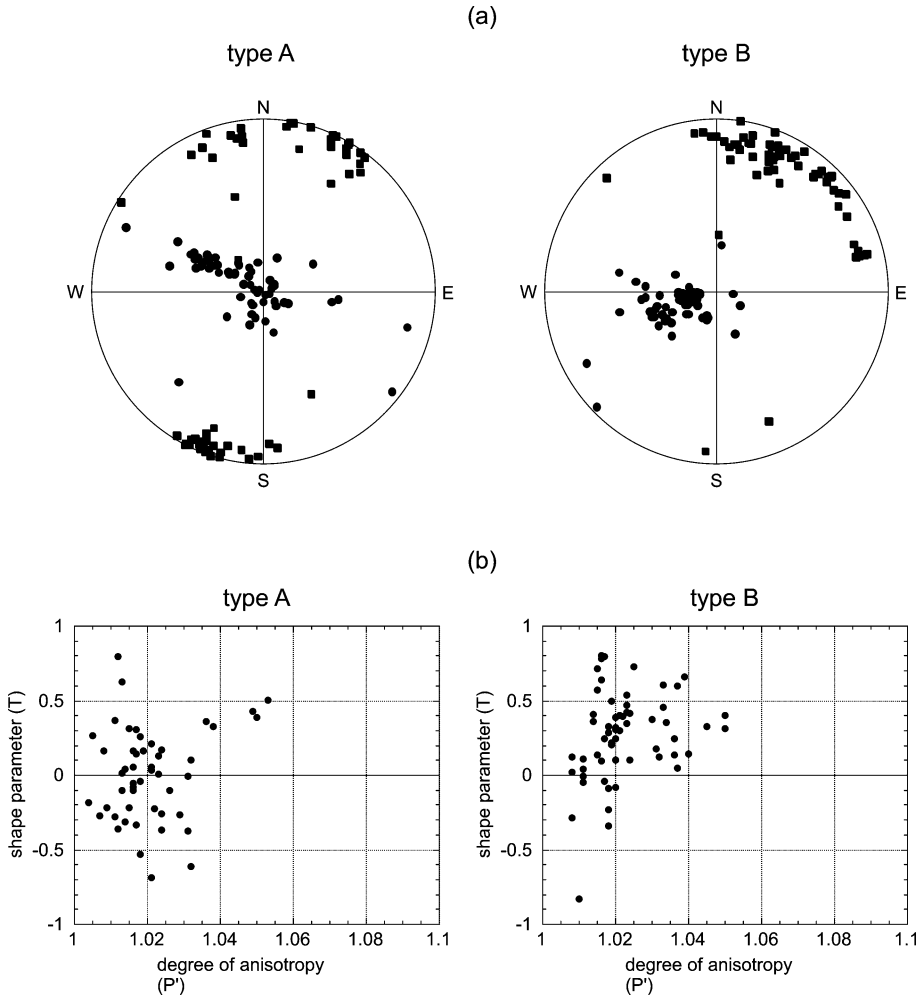


Figure 5. (a) Orientations of the maximum and minimum principal magnetic susceptibility axes for types A and B (after tilt correction). Solid squares and circles represent maximum and minimum axes, respectively. (b) Plot of shape parameter T versus degree of anisotropy P' (Jelínek 1981) for types A and B.

can produce inclination shallowing (Garcés *et al.* 1996) and tilting of the maximum AMS axis toward the sourceward direction (Rees & Woodall 1975).

We observed the sandstone samples from this locality under an optical microscope in order to examine the existence of imbricated structures which form in association with aqueous flow. Polished thin sections were cut perpendicular to the bedding plane along a northeast–southwest direction that corresponds to the direction of the maximum AMS axis, since the dip direction of the maximum AMS axis is expected to approximate the sourceward direction of the flow. Examinations of the polished sections indicate that sedimentary particles have a preferred direction (Fig. 6). A northeastward-dipping imbricated structure of biotite is readily observed. The imbricated structure is exclusively observed in type B samples and not observed in those of type A. This microscopic aspect indicates the deposition from currents in the type B locality and is considered relevant to the shallowing effect.

4 DISCUSSION

4.1 South China Affinity of the Kimpo Area

An Early Jurassic palaeomagnetic pole for the North China Block is located at 81.5°N, 302.5°E with $A_{95} = 5.5°$ (Uno & Huang 2003) (Table 2). This pole, together with the Late Triassic one by Yang *et al.* (1991), constitutes a large segment in the Mesozoic apparent polar wander path for the North China Block (Fig. 7). Early Jurassic and Early–Middle Jurassic poles for the South China Block lie at 60.4°N, 204.5°E with $A_{95} = 14.0°$ (Gilder *et al.* 1993) and 79.9°N, 221.8°E with $A_{95} = 5.3°$ (Yang & Besse 2001), respectively. A large Mesozoic segment for this block is also observed from poles between Middle Triassic and Early–Middle Jurassic times (Fig. 7). It is apparent that the pole positions for the North and South China Blocks approach to each other through time. The sense of polar motion for the two blocks suggests counter-clockwise rotation of the North China Block and clockwise rotation of the South China Block. The Early Jurassic pole for the North China Block (Uno & Huang 2003) and the Early–Middle Jurassic one for the South China Block (Yang & Besse 2001) reveal a significant separation, suggesting that relative movement between the two blocks was not completed by the Early Jurassic. The process of relative movement eventually ceased during the Middle–Late Jurassic (Gilder & Courtillot 1997) or the Early Cretaceous (Yokoyama *et al.* 2001).

An Early Jurassic palaeomagnetic pole for the Kimpo area is calculated on the basis of the new data set of primary magnetization. It is located at 54.1°N, 230.2°E with $A_{95} = 4.1°$. The pole for the Kimpo area falls closer to the Jurassic poles for the South China Block than that of the North China Block (Fig. 7), suggesting that the Kimpo area, as well as the South China Block, has experienced clockwise

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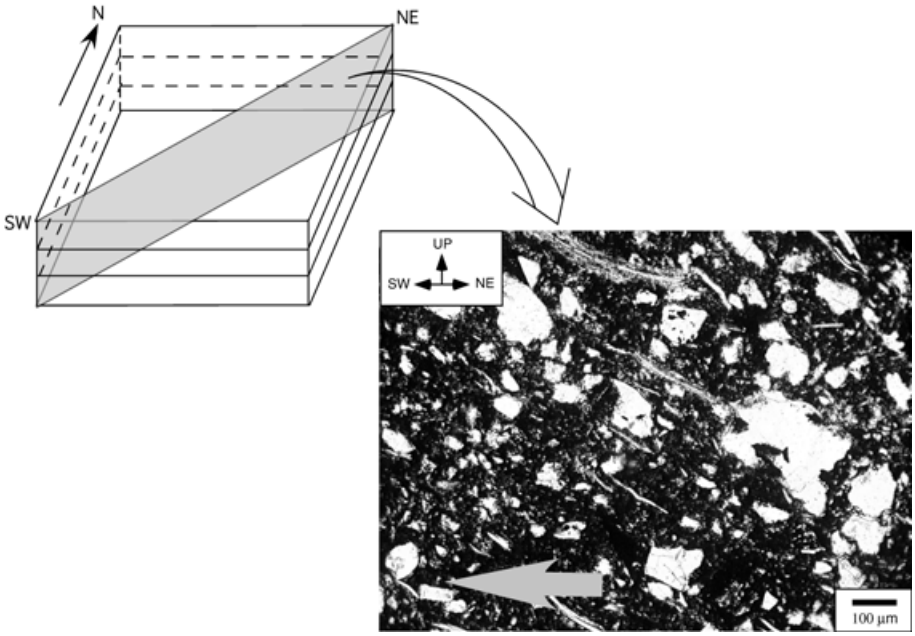


Figure 6. Illustration showing relationship between bedding planes and microstructural features for sample MJ906 (type B). Photomicrograph, taken in plane-polarized light, is a vertical section to bedding planes along a northeast–southwest direction, which corresponds to the direction of the maximum AMS axis. The arrow indicates the palaeocurrent direction inferred from the sourceward-dipping imbricated structure of biotite. Scale bar is 100 μm.

Table 2. Palaeomagnetic poles for the Kimpo area, North China Block and South China Block.

Age	$N(n)$	Pole			Reference
		Lat. (°N)	Long. (°E)	A_{95}(°)	
Kimpo area					
J1	(11)	54.1	230.2	4.1	This study
Tr3	(14*)	39.6	207.5	7.0	Kim *et al.* (1993)
North China Block					
J1	(15)	81.5	302.5	5.5	Uno & Huang (2003)
Tr3	(11)	62.3	7.7	3.8	Yang *et al.* (1991)
Tr2	7	61.0	2.0	2.9	Yang & Besse (2001)
South China Block					
J1-2	5	79.9	221.8	5.3	Yang & Besse (2001)
J1	3	60.4	204.5	14.0	Gilder *et al.* (1993)
Tr2	8	42.7	212.6	6.0	Yang & Besse (2001)

Key: Tr2, Middle Triassic; Tr3, Late Triassic; J1, Early Jurassic; J1-2, Early to Middle Jurassic; J2, Middle Jurassic; $N(n)$, the number of studies (sites/samples*); Lat., north latitude; Long., east longitude; A_{95}, radius of the circle of 95 per cent confidence.

rotation with respect to the North China Block during the collision event. It would appear that the Kimpo area has been subjected to rotational motion as an integral part of the South China Block. This correlation between the South China Block and the Kimpo area in terms of rotation requires a direct geographical connection at the time of collision. Palaeontological observations reveal that the Triassic *Dictyophyllum–Clathropteris* floras, which are found in a wide area of the South China Block, are also observed in the Kimpo area (Otoh & Sasaki 1998), corroborating the geographical continuity between the South China Block and the Kimpo area.

A Late Triassic pole for the Kimpo area (39.6°N, 207.5°E with $A_{95} = 7.0°$ (Kim *et al.* 1993) further supports the clockwise sense of rotation during the collision event. This pole together with the Early Jurassic one define a Late Triassic–Early Jurassic segment of the apparent polar wander path for the Kimpo area, and the segment falls within the Middle Triassic–Early Jurassic segment for the South China Block. This observation from palaeomagnetic poles suggests that the South China Block and the Kimpo area were in continuity at the time of collision, leading to the consistent clockwise sense of rotation during the collision.

4.2 Tectonic affinity between the Korean Peninsula and the Chinese blocks

The Nangrim and Ryeongnam Blocks are situated in the northern and southeastern parts respectively of the Korean Peninsula (Fig. 1b). They are regarded as the eastern extension of the North China Block. The Nangrim Block is a part of the North China Block because of a geographically and geologically direct connection between the two regions. The Ryeongnam Block is also determined to be a part of the North China Block from palaeomagnetic data.

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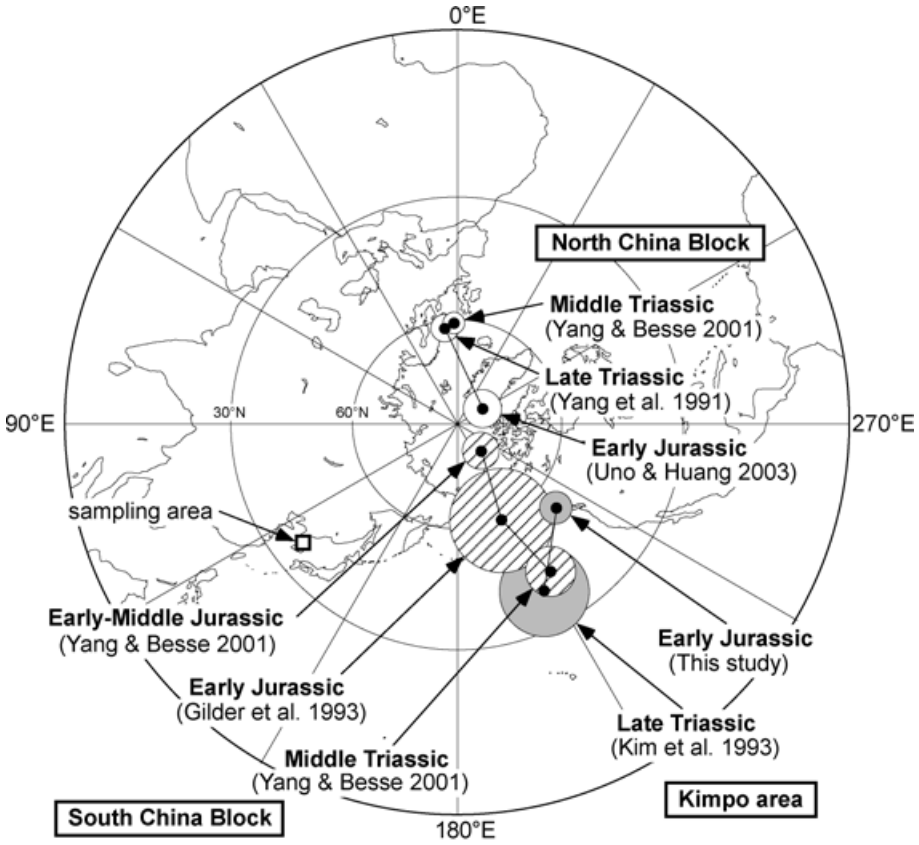


Figure 7. Triassic and Jurassic palaeomagnetic poles (with associated 95 per cent confidence limits) for the Kimpo area, North China Block and South China Block. The poles for the Kimpo area show shaded confidence circles. The poles for the North (South) China Block show the open (striped) confidence circles. The open square (sampling area) denotes the reference point of the Kimpo area. Refer to Table 2 for data details.

No significant palaeolatitude contrast exists between the observed palaeolatitude and the expected one derived from the poles for the North China Block through Palaeozoic times (Lee *et al.* 1997), and Late Palaeozoic–Early Mesozoic palaeomagnetic poles for the Ryeongnam Block agree well with coeval poles for the North China Block (Doh & Piper 1994; Uno 1999).

The present study indicates that the Korean Peninsula has an area that behaved as an integral part of the South China Block during the collision (i.e. the Kimpo area), suggesting that the Korean Peninsula is composed of areas of both North and South China affinities. Two interpretations have been proposed regarding South China tectonic elements in the Korean Peninsula. On the one hand, Li (1994) and Uno (1999) suggest that the basement rocks of the central part of the Korean Peninsula (Gyeonggi Block) are derived from the South China Block. This interpretation implies that the Gyeonggi Block, as the eastern part of the South China Block, was obducted onto the North China Block at the site of the present Korean Peninsula. On the other hand, Chang (1995) and Otoh *et al.* (1999) suggest that all the basement rocks comprising the Korean Peninsula are of North China origin. They interpret that the Imjingang Belt, located between the Nangrim and Gyeonggi Blocks (Fig. 1b), has South China affinity and is the allochthonous unit on the Korean Peninsula. Although both of the tectonic interpretations agree that the Korean Peninsula suffered obduction of the South China Block around the central part of the peninsula, the affinity of the Gyeonggi Block is not consistent.

Basement rocks of the Gyeonggi Block provide the depleted mantle Nd model ages (T_{DM}) that are comparable with those from the Ryeongnam Block (Lan *et al.* 1995). Cheong *et al.* (2000) also pointed out common Archaean to Proterozoic basement ages between the Gyeonggi and Ryeongnam Blocks on the basis of the T_{DM} values. A good linearity in $^{207}Pb/^{204}Pb$ versus $^{206}Pb/^{204}Pb$ plots is observed between the Gyeonggi and Ryeongnam Blocks, yielding an apparent basement age of about 2.0 Ga (Cheong *et al.* 2000). These isotopic data suggest that the basement rocks in the Gyeonggi Block are similar to those in the Ryeongnam Block that is correlated with the North China Block.

A recent palaeontological study of Cambrian conodonts suggests a faunal similarity between the Ryeongnam and Gyeonggi Blocks. Jeong & Lee (2000) pointed out that the conodonts from the Ryeongnam Block are well correlated with those from the North China Block and that no remarkable faunal difference is observed between the Ryeongnam and Gyeonggi Blocks. Recent palaeontological and isotopic data supporting a close connection between the Ryeongnam and Gyeonggi Blocks argue that the Gyeonggi Block is probably a part of the North China Block. Therefore, we prefer the latter tectonic interpretation by Chang (1995) and Otoh *et al.* (1999), who claim that all the basement rocks comprising the Korean Peninsula are of North China origin.

4.3 South China tectonic elements in the Korean Peninsula

The Imjingang Belt possesses Middle Ordovician to Early Carboniferous strata that are hardly observed in the eastern half of the North China Block. This belt has Silurian and Devonian fossils that are correlated with those from the South China Block.

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The Silurian Imjin Group yields *Leptophloeum*, and the Devonian Hwangju Group yields *Rhabdotetradium*, *Agetolites* and *Palaeofavosites* (Chang 1995; Otoh *et al.* 1999). It is therefore considered that the Palaeozoic strata in the Imjingang Belt are derived from the South China Block.

The Kimpo sediments of the present study belong to the upper part of the Daedong Supergroup. This supergroup unconformably overlies the Silurian strata of the Imjingang Belt (Kobayashi 1935) and is interpreted to constitute a part of the belt (Otoh & Sasaki 1998, Fig. 1b). It is therefore concluded that the Imjingang Belt, including the Kimpo area, is a part of the South China Block on the Korean Peninsula. These South China elements were obducted onto the North China Block at the site of the present Korean Peninsula during the welding event, and hence the Korean Peninsula is composed of areas of both North and South China affinities.

The Korean Peninsula has been involved in the collision and suturing event of the North and South China Blocks as their easternmost parts. The palaeomagnetic results of the present study show that the mode of the North–South China collision in the easternmost part, around the Korean Peninsula, is also explained by the scissor-like process (Zhao & Coe 1987; Gilder & Courtillot 1997) which predicts clockwise rotation of the South China elements during the collision.

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